

AI-Proctored Secure Examination System

Project Technical Documentation & Analysis

1. Project Overview

The AI-Proctored Secure Examination System is a web-based client-side examination portal designed to ensure academic integrity during online tests. It monitors user behaviors using browser-native capabilities and client-side artificial intelligence models (machine learning object detection and custom computer vision algorithms).

The application is structured into two main components:

- **Exam Portal (Client Interface):** The interface where the candidate registers, completes verification, and takes the exam.
- **HR Admin Portal (Dashboard):** The real-time control panel where the administrator configures exam parameters, views metrics, and inspects violation logs and screenshots.

2. Technology Stack

The application is designed using a lightweight, performant, and serverless client-side architecture to bypass heavy backend infrastructure requirements.

Component	Technology	Description
Core Structure	HTML5 / CSS3 / Vanilla JS	Native web components without heavy frameworks for maximum loading speed.
Build & Bundler	Vite	Modern, fast front-end tooling to bundle modules and compile static assets.
Data Layer	LocalStorage	Used for cross-tab syncing and state persistence between the exam tab and HR portal.
AI Processing	TensorFlow.js & COCO-SSD	Runs client-side object detection models in real-time in the browser.
Audio Processing	Web Audio API	Microphone audio amplitude analysis to detect vocal noise without server transfers.
Screen Share	MediaDevices API	Native screen capture to verify full-desktop share and screenshot restrictions.

3. AI Models & Computer Vision Algorithms

A. TensorFlow.js COCO-SSD Model

- **Model Type:** MobileNet v2 (lightweight backbone) trained on the COCO dataset.
- **Execution:** Runs client-side directly in the candidate's browser via a background loop.
- **Object Monitoring:** Detects candidate absence (person count is 0), secondary persons (multiple faces detected), and mobile phones (immediate lockout trigger).

B. Custom Eye-Gaze Tracking Engine

- **Method:** Pupil-Iris Pixel Luminance Analysis.
- **Execution:** Captures frames from the webcam at 500ms intervals, scales them down to 160x120, and runs pixel-by-pixel brightness scans.
- **Gaze Direction Classification:** Detects eye gaze centering, left/right deviations (horizontal pixel shift > 15), and look-down deviation (vertical pixel shift > 7.5).

4. Implemented Features

- **Dynamic Question-Level Timings**

Added a timeLimit parameter to all questions. Extended the HR Portal modal with a 'Time Limit (seconds)' field, showing a stopwatch badge (e.g. ■■ 300s) next to each question on the list. The candidate's exam timer calculates the total exam duration dynamically by summing the time limits of all active questions.

- **Looking Down Gaze Detection & Warning System**

Implemented vertical pupil tracking. Added a continuous looking-down tick counter. To prevent false positives, a warning is only sent after 2 minutes (240 ticks) of continuous down-gaze. Subsequent warnings are fired every 30 seconds (60 ticks) up to a hard cap of 5 warnings. Looking back at the screen resets the timer.

- **Conditional Lockouts (Warning-Only Mode)**

Changed the lockout policy so that candidates are only locked out of the exam when a mobile phone (cell_phone) is detected by the AI. All other violations (tab switches, looking left/right/down, loud noise, face absent) issue warnings but allow the candidate to continue the exam.

- **Gaze Violation Screenshot Captures**

Added a canvas snapshot helper that captures a frame from the candidate's webcam feed upon looking away or down. Compresses the image as a lightweight JPEG Data URL (~4KB) with a timestamp watermark and displays it inside the HR portal drawer.

5. Pros and Cons of Browser-Client Proctoring

Pros / Advantages	Cons / Limitations
• Absolute Privacy: All processing remains local in the browser; no video/audio uploaded. • Zero Server Cost: No GPU cloud servers needed, making system highly scalable. • Low Bandwidth: Operates on slow connections since video data is not streamed. • No Installs: Works instantly in any modern web browser without plugins.	• Client CPU Load: Machine learning runs locally, which can cause lag on older machines. • Evasion Vulnerability: Candidate can physically close the browser window. • Storage Capacity Limit: LocalStorage restricts total session data (with images) to a 5MB limit. • Lighting Sensitivity: Gaze tracking accuracy depends on camera quality and ambient light.

6. Future Recommendations

- **IndexedDB Migration:** Move session logs from LocalStorage to IndexedDB to eliminate the 5MB storage limit and allow unlimited screenshot storage.
- **Server Syncing Integration:** Connect the client-side data layer to a secure backend database (e.g. Firebase, Node.js) for persistent remote reporting.
- **Face Mesh Model:** Upgrade the gaze tracking engine to a lightweight Mediapipe Face Mesh model to track eye movement with sub-pixel precision under varied lighting conditions.